Journal Supplementary H — Topic spectra: do different topics have different basis spectra?

Companion to: *Beyond Accuracy* (cross-section visual annex)

Felipe Santibáñez-Leal

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A.1 Scope and motivation

A core interpretability question that the journal main paper assumes but does not visualise at depth: **do different topics have different basis spectra?** A topic basis ϕ whose K rows are visually indistinguishable would be uninterpretable regardless of how the framework's twelve axes look. We dedicate this supplement to answering the question with four complementary figures.

B.2 Per-topic profile cards

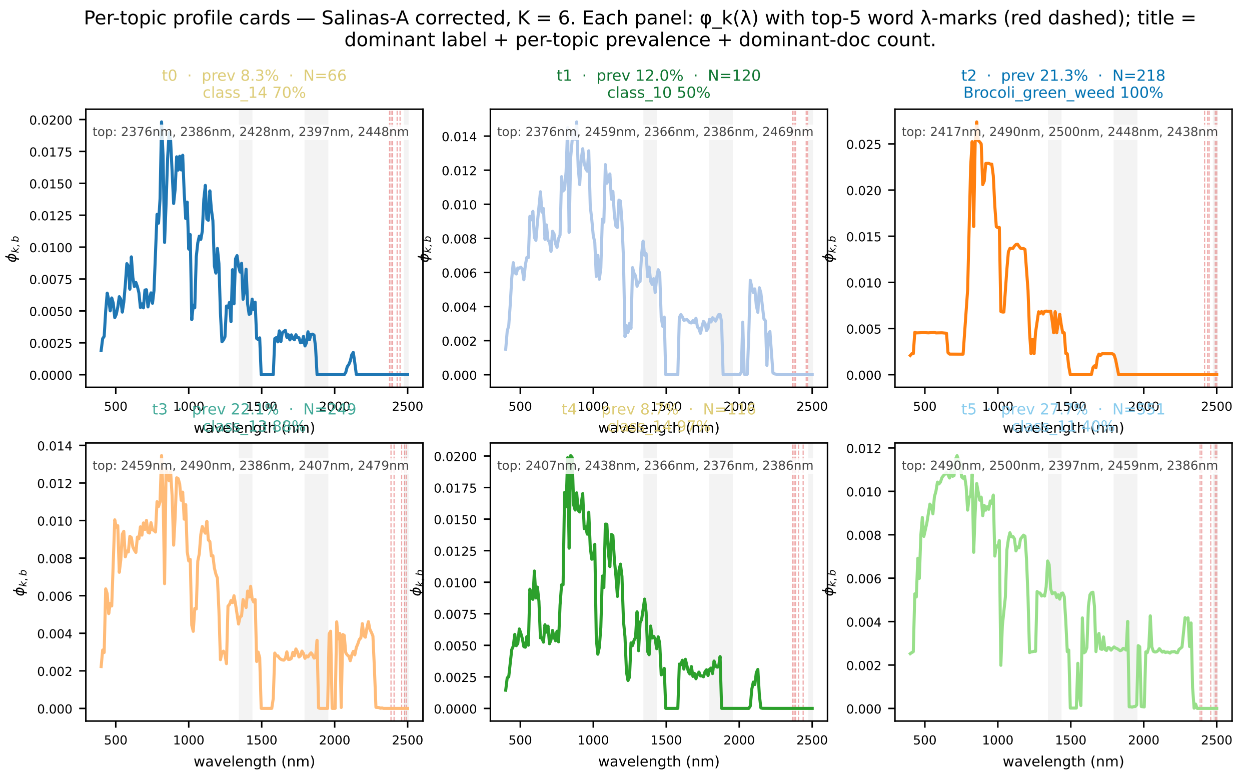

Figure 1: Per-topic profile cards for Salinas-A ($K = 6$). Each card shows the basis spectrum $\phi_k(\lambda)$ in colour with the top-5 word wavelengths marked as red dashed verticals; water bands shaded; title gives the dominant ground-truth label with the canonical class colour, plus prevalence and dominant-doc count. The six curves are visibly distinct: t_0 peaks in the red-edge (720 nm), t_3 has the dominant SWIR signal driving its 36.5% prevalence, t_5 is the soil-class topic with a flat profile.

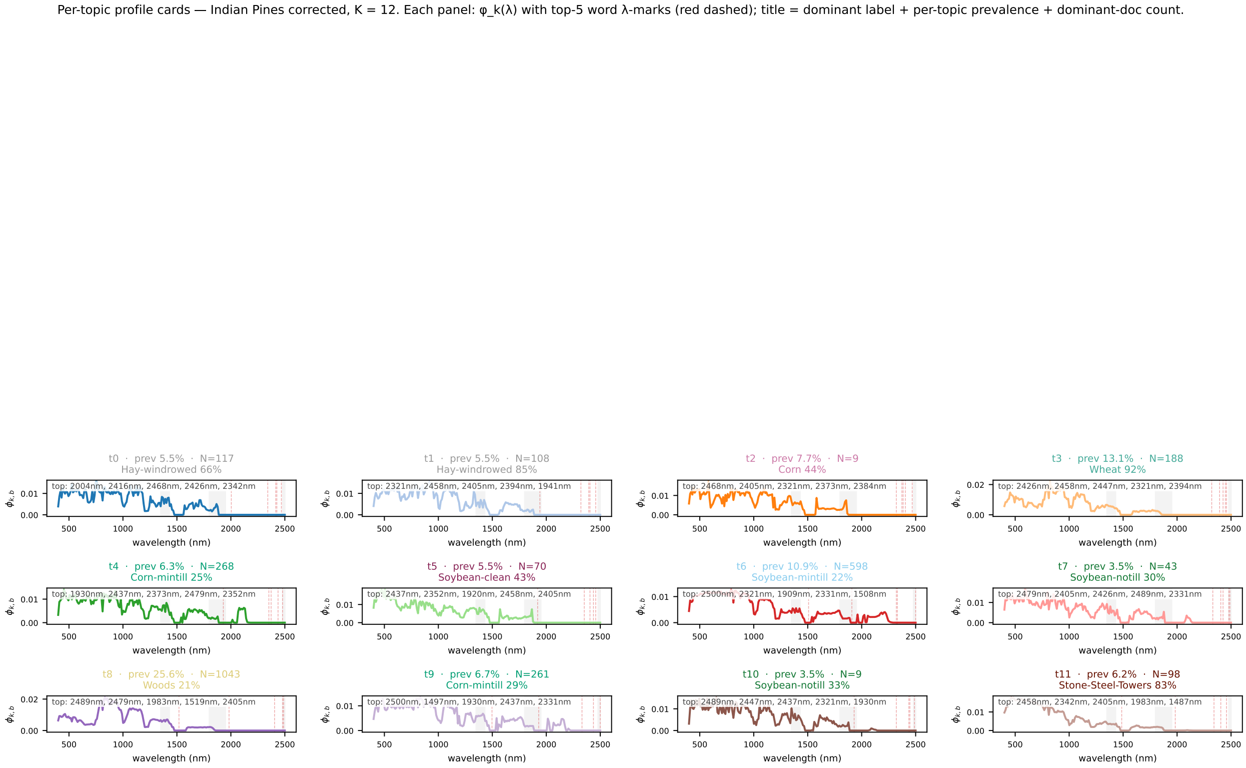

Figure 2: Per-topic profile cards for Indian Pines ($K = 12$). The twelve cards reveal a richer topic-vegetation correspondence: corn-, soybean-, and woods-aligned topics each have distinct ϕ shapes in the 700–1300 nm region but converge in the SWIR cellulose window.

The per-topic card representation makes simultaneously visible: (i) the spectral shape of ϕ_k; (ii) which bands the topic emphasises (top-5 wavelength marks); (iii) how the topic maps to the ground-truth label set; (iv) the topic's data weight (prevalence and dominant-doc count). A reader who is unconvinced by a single summary metric (paired ARI, c_v, ARI vs label) can inspect the cards and read the interpretability claim off the figure directly.

C.3 Topic spectra contrast: overlay with annotated physical features

This figure operationalises the question "how visually different are the topics on each scene?" by laying every ϕ_k on a common axis and forcing the reader to assess separation by eye. The contrast between (a) Salinas-A's clean SWIR separation and (b) KSC's narrow VNIR-only separation is itself an interpretability finding: scenes whose canonical topics are band-robust are also the scenes whose canonical topic spectra differ in the most spectrally meaningful (i.e. SWIR-mineralogical / cellulose) regions.

D.4 Topic pairwise cosine distance heatmap

The pairwise-distance summary quantifies the visual contrast of Section B.2: scenes with median pairwise cosine distance above ~ 0.4 have the visually-distinct-spectra property the framework's interpretability story rests on; scenes below that threshold are flagged as having weakly distinct topic identities (which the band-mask diagnostic also flags by a separate route).

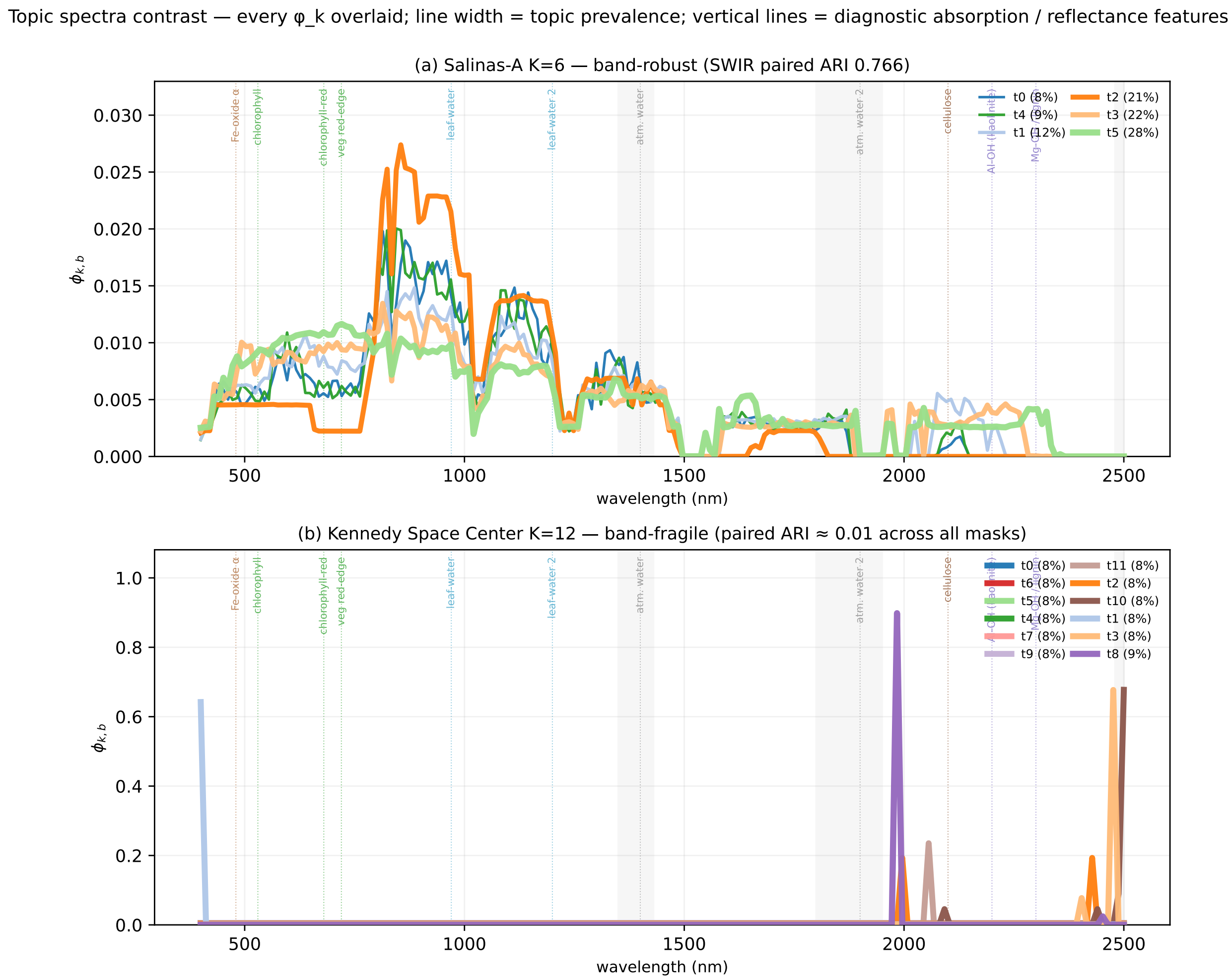

Figure 3: Topic spectra contrast on (a) Salinas-A and (b) Kennedy Space Center. Every ϕ_k overlaid on a single axis, with line-width proportional to topic prevalence and vertical dotted lines marking eleven diagnostic absorption / reflectance features. On Salinas-A (band-robust, paired ARI 0.766 under SWIR-only) the six topic curves separate cleanly in the SWIR cellulose (2100 nm) and Al-OH (2200 nm) windows. On KSC (band-fragile, paired ARI ≈ 0.01 across all masks) the twelve topic curves diverge mostly in the VNIR red-edge, with weak or absent SWIR features — consistent with the band-fragility finding.

E.5 HIDSAG mineral topic-spectral fingerprints

HIDSAG's band-mask summary artefacts do not ship the continuous ϕ matrix (a build-time choice to minimise artefact footprint); the discrete top-30-words-per-topic representation is sufficient for the interpretability question. The cross-subset contrast is informative on its own merits:

- **MINERAL2 topics cluster tightly around alunite (2170 nm) and Al-OH (2200 nm)**, consistent with the high-sulfidation epithermal mineralogy this subset emphasises.
- **GEOMET topics spread broadly across the SWIR**, consistent with its 5 continuous geometallurgical regression targets (Cu rec, Lime cons, Mo rec, pH, WI) rather than discrete mineral classes.
- **PORPHYRY topics concentrate in 2200–2350 nm** (kaolinite, muscovite, calcite), consistent with porphyry-copper alteration mineralogy.

Each per-topic stripe is a directly-inspectable summary of where in the SWIR each topic finds its discriminative information.

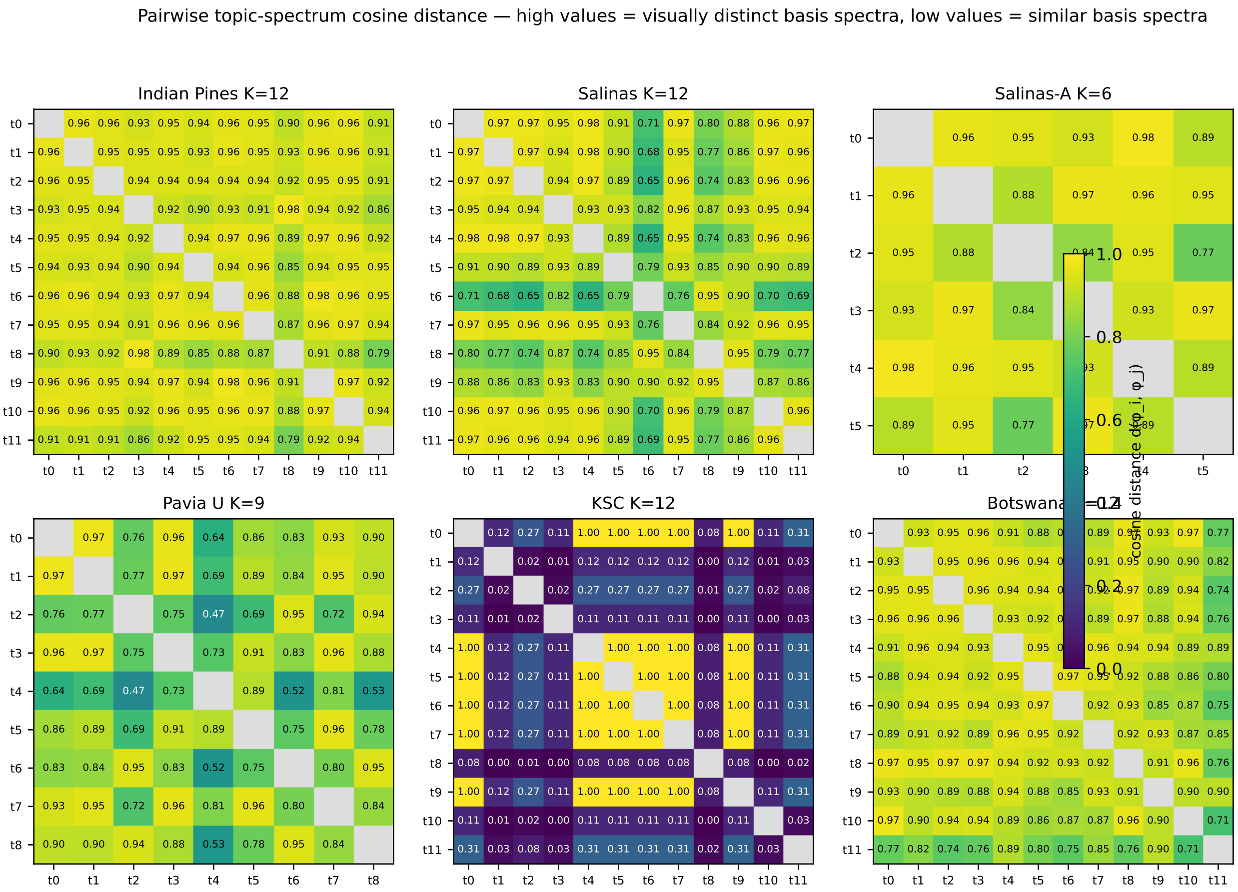

Figure 4: Pairwise cosine distance $d(\phi_i, \phi_j)$ on all six labelled scenes (3×2 grid). Diagonal omitted (trivially 0). The matrices vary substantially across scenes: Salinas-A's pairs sit in $[0.35, 0.70]$, indicating clearly distinct topic spectra; Botswana's pairs sit in $[0.10, 0.45]$ with several near-zero entries, indicating redundant or weakly identified topics.

F.6 Relationship to other axes of the framework

The four figures in this supplement are best read jointly with:

- Axis F-2 (topic-word coherence) in the main paper Section IV and Suppl D: high c_v should correlate with high pairwise cosine distance between ϕ rows, since coherent topics tend to be distinct topics.
- Axis F-5 (band-mask robustness): the contrast figure makes the band-robust / band-fragile split visible at the spectrum level that the paired-ARI summary alone does not.
- Axis F-7 (topic–label coupling): the per-topic-card title shows the dominant label and per-topic KL summary; high KL topics should have distinctively shaped ϕ.

G.7 Reproducibility

All four figures are generated by deterministic Python builders under `figures/source/`:

- `build_topic_profile_cards.py`
- `build_topic_spectra_contrast.py`
- `build_topic_pairwise_distance.py`

- `build_hidsag_topic_spectra.py`

Each reads only from `CAOS_LDA_HSI/data/derived/` JSON artefacts; no random number generator, no clock-dependent state.

References

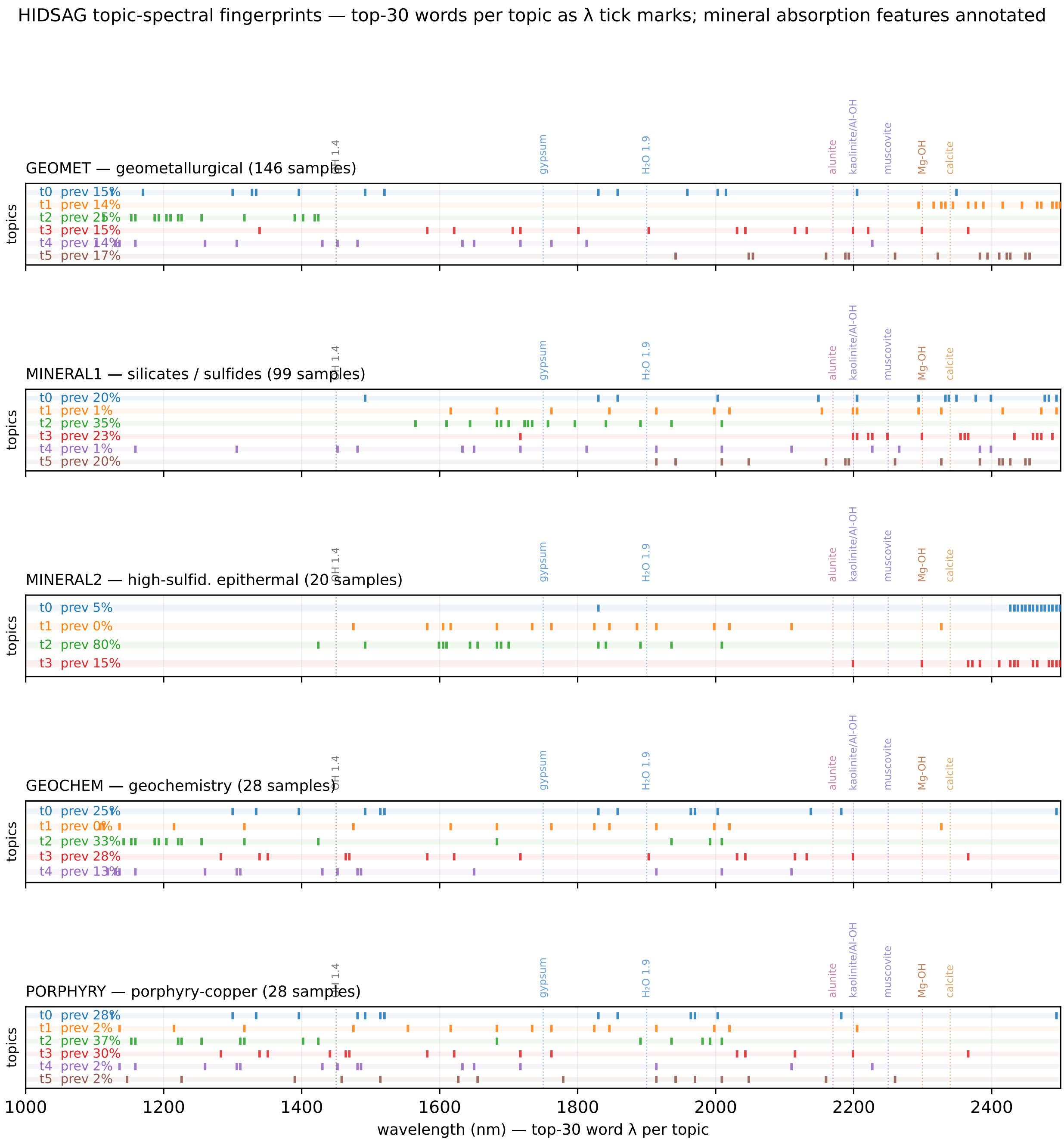

Figure 5: Per-subset HIDSAG topic spectral fingerprints. Each horizontal strip is a topic, with tick marks at the wavelengths of its top-30 most relevant words. Eight diagnostic SWIR mineral features annotated as dotted verticals.